

Tarea 13

5.43) 24.4 L
0.280

$$M = 0.400 \text{ g} \left| \frac{1.25 \times 10^{-3}}{1} \right| = 320 \text{ g/mol}$$

5.45) $1 \times 10^{-3} \text{ atm} \times 1 \text{ L} = n \times 0.0821 \text{ L} \cdot \text{atm} / \text{mol} \cdot \text{K} \times 250$

$$n = \frac{1 \times 10^{-3} \text{ atm} \times 1 \text{ L}}{0.0821 \text{ L} \cdot \text{atm} / \text{mol} \cdot \text{K} \times 250} = 4.87 \times 10^{-5}$$

$$4.87 \times 10^{-5} \left| \frac{6.023 \times 10^{23} \text{ mol}^{-1}}{1 \text{ mol}^{-1}} \right| = 2.93 \times 10^{19} \text{ mol} / \text{O}_3$$

5.47) a) $d = \frac{(1 \text{ atm} \cdot \text{L} / \text{mol} \cdot \text{K}) (P_m)}{0.0821 \cdot 300.15 \text{ K}} = \frac{(1 \text{ atm} \cdot \text{L} / \text{mol} \cdot \text{K}) (0.397 \text{ g/L})}{0.0821 \cdot 300.15 \text{ K}}$

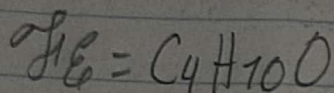
$$= 0.016 \text{ g/cm}^3 \left| \frac{1 \text{ cm}^3}{0.001 \text{ L}} \right| = 16 \text{ g/L}$$

b) $n = \frac{(1 \text{ atm}) (2.1 \text{ L})}{0.0821 \cdot 300.15} = 0.85 \text{ mol}$ $PM = 0.85 \text{ mol} \cdot 4.65 \text{ g} = 0.40 \text{ mol/g}$

$$5.49) 64.9 \text{ g C} \left| \frac{1 \text{ mol}}{12.011 \text{ g}} \right| = 5.40 \quad 4$$

$$13.5 \text{ g H} \left| \frac{1 \text{ mol}}{1.00794} \right| = 13.61 \quad 10$$

$$21.6 \text{ g O} \left| \frac{1 \text{ mol}}{15.9994} \right| = 1.35 \quad 1$$



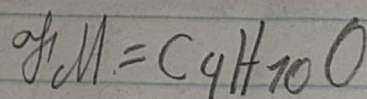
$$m = 74.12 \text{ g/mol}$$

$$M = \frac{2.3 \text{ g} \times 0.0821 \times 393.15}{0.99 \times 1}$$

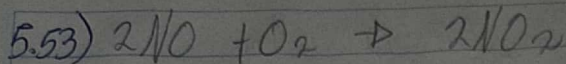
$$P = 750/760 = 0.99 \text{ atm}$$

$$M = 75 \text{ g/mol}$$

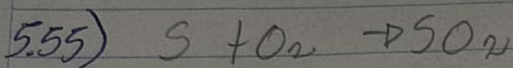
$$74.12/75 = 1$$



5.51)



$$\begin{array}{c|c|c|c} 9\text{L NO} & 1\text{mol NO} & 2\text{mol NO}_2 & 30.07\text{L NO}_2 \\ \hline & 30.07\text{L NO} & 2\text{mol NO} & 1\text{mol NO}_2 \end{array} = 9\text{L NO}_2$$



$$\begin{array}{c|c|c|c|c|c|c} 2.54\text{Kg S} & 1000\text{g S} & 1\text{mol SO}_2 & 22.4\text{L} & 30.5+273 & 1\text{L} & 1000\text{ml} \\ \hline & 1\text{Kg S} & 32.06\text{g S} & 1\text{mol SO}_2 & 273 & 1.2\text{atm} & 1\text{L} \end{array} =$$

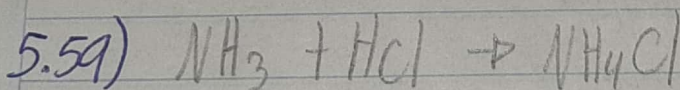
$$1,643,810.50\text{ml SO}_2$$

$$5.57) \text{MdP} = 30.97376$$

$$F = 18.99840$$

$$2(30.97376) + 4(18.99840) = 137.94$$

$$P_2 F_4$$

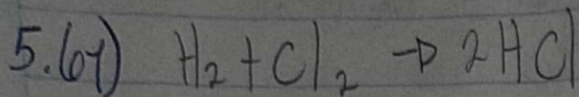


$$\begin{array}{c|c} 73\text{g NH}_3 & 1\text{mol NH}_3 \\ \hline & 17.03\text{g NH}_3 \end{array} = 4.29\text{mol NH}_3$$

$$4.29 - 2 = 2.29\text{mol NH}_3$$

$$\begin{array}{c|c} 73\text{g HCl} & 1\text{mol HCl} \\ \hline & 36.46\text{g HCl} \end{array} = 2\text{mol HCl}$$

$$2.29 \text{ g/mol NH}_3 \left| \begin{array}{c} 22.4 \text{ L} \\ 1 \text{ mol NH}_3 \end{array} \right| \begin{array}{c} 287 \\ 273 \end{array} \left| \begin{array}{c} 760 \\ 752 \end{array} \right| = 54.50 \text{ L NH}_3$$

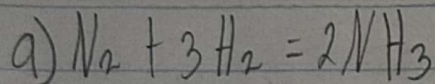


$$(1)(5.6) = n(0.0821)(273)$$

$$n = 0.25 \text{ mol H}_2$$

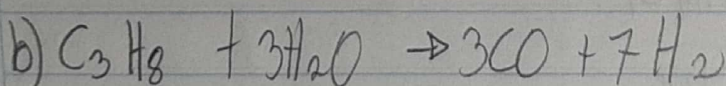
$$0.25 \left| \begin{array}{c} 2 \text{ mol HCl} \\ 1 \text{ mol H}_2 \end{array} \right| \left| \begin{array}{c} 36.46 \text{ g HCl} \\ 1 \text{ mol HCl} \end{array} \right| = 18.23 \text{ g HCl}$$

5.63)



$$12.8 \text{ L NH}_3 \left| \begin{array}{c} 1 \text{ mol NH}_3 \\ 16.02 \text{ L NH}_3 \end{array} \right| \left| \begin{array}{c} 1 \text{ mol N}_2 \\ 2 \text{ mol NH}_3 \end{array} \right| \left| \begin{array}{c} 16.02 \text{ L N}_2 \\ 1 \text{ mol N}_2 \end{array} \right| = 6.4 \text{ L N}_2$$

$$12.8 \text{ L NH}_3 \left| \begin{array}{c} 1 \text{ mol NH}_3 \\ 16.02 \text{ L NH}_3 \end{array} \right| \left| \begin{array}{c} 3 \text{ mol H}_2 \\ 2 \text{ mol NH}_3 \end{array} \right| \left| \begin{array}{c} 16.02 \text{ L H}_2 \\ 1 \text{ mol H}_2 \end{array} \right| = 19.2 \text{ L H}_2$$



$$8.96 \text{ L H}_2 \left| \begin{array}{c} 1 \text{ mol H}_2 \\ 22.4 \text{ L H}_2 \end{array} \right| \left| \begin{array}{c} 1 \text{ mol C}_3\text{H}_8 \\ 10 \text{ mol H}_2 \end{array} \right| \left| \begin{array}{c} 22.4 \text{ L C}_3\text{H}_8 \\ 1 \text{ mol C}_3\text{H}_8 \end{array} \right| = 0.90 \text{ L C}_3\text{H}_8$$

$$8.96 \text{ L H}_2 \left| \begin{array}{c} 1 \text{ mol H}_2 \\ 22.4 \text{ L H}_2 \end{array} \right| \left| \begin{array}{c} 6 \text{ mol H}_2\text{O} \\ 10 \text{ mol H}_2 \end{array} \right| \left| \begin{array}{c} 22.4 \text{ L H}_2\text{O} \\ 1 \text{ mol H}_2\text{O} \end{array} \right| = 53.76 \text{ L H}_2\text{O}$$